

# Blocco 3

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**Warning.** In the following we will adopt the following conventions:

- In each exercise,  $P$  will denote the poset used.
- When something coincide when seen in  $V$  or in  $V[G]$  we we'll not specify where we are looking at it. So for example we we'll always write  $\omega$  instead of  $\omega^V$ ,  $\omega^{V[G]}$ .
- We won't specify if something is considered in  $V$  or in  $V[G]$  if it is clear from the context.

For the next problem you can use that for inaccessible  $\lambda$ ,  $Col(\omega_1, < \lambda)$  has the  $\lambda - cc$  and if  $G$  is a  $Col(\omega_1, < \lambda)$ -generic filter over the ground model  $V$ , then  $V[G] \models \lambda = \omega_2$ .

**Esercizio 1.** Suppose that  $\lambda$  inaccessible. Show that  $Col(\omega_1, < \lambda)$  forces that CH holds.

*Proof.* Since  $\omega < \lambda$ , we have by inaccessibility of  $\lambda$  that  $|2^\omega| < \lambda$ . Let  $G$  be a generic filter for  $P$ . First we claim that, since  $P$  is  $\omega_1$ -closed, we have  $(2^\omega)^V = (2^\omega)^{V[G]}$ .

- $(2^\omega)^V \subseteq (2^\omega)^{V[G]}$ : because  $V \subseteq V[G]$ .
- $(2^\omega)^V \supseteq (2^\omega)^{V[G]}$ : Let  $V[G] \models A \subseteq \omega$ . We want to show that  $A \in V$ . Let  $\dot{A}$  be a  $P$ -name over  $V$  for  $A$ . Now consider  $\langle p_n : n < \omega \rangle$ , defined inductively as follows:

1.  $n = 0$ :

- if  $V[G] \models 0 \in A$  take  $p_0 \in G$  such that  $p_0 \Vdash \check{0} \in \dot{A}$
- if  $V[G] \models 0 \notin A$  take  $p_0 \in G$  such that  $p_0 \Vdash \check{0} \notin \dot{A}$

2.  $n \Rightarrow n + 1$ :

- if  $V[G] \models n + 1 \in A$  take  $p_{n+1} \in G, p_{n+1} \leq p_n$  such that  $p_{n+1} \Vdash n \check{+} 1 \in \dot{A}$
- if  $V[G] \models n + 1 \notin A$  take  $p_{n+1} \in G, p_{n+1} \leq p_n$  such that  $p_{n+1} \Vdash n \check{+} 1 \notin \dot{A}$

Now let  $\bar{p} \in G, \bar{p} \leq p_n$  for all  $n < \omega$ . Such an element exists because one can consider  $p' = \bigcup_n p_n, D = \{q : q \leq p'\} \cup \bigcup_n \{q : q \perp p_n\}$ . We claim that  $D$  is dense. In fact, let  $r \in P$ . If  $r \perp p_n$  for some  $n < \omega$  we already have  $r \in D$ . So we can assume  $r \not\perp p_n$  for all  $n < \omega$ . Define  $\langle r_n : n < \omega \rangle$  as follows:

1.  $n = 0$ : take  $r_0 \leq r, p_0$

2.  $n \Rightarrow n + 1$ :

- if  $r_n \perp p_{n+1}$  then we have  $r_n \in D, r_n \leq r$  so one can stop.
- othewise take  $r_{n+1} \leq r_n, p_{n+1}$ .

If the sequence stopped we proved the density, so assume the sequence did not stop. In this case take  $r' = \bigcup_n r_n$ . Then we have  $r' \leq p_n$  for all  $n < \omega$ , which implies  $r' \leq p'$ . So we have  $r' \in D, r' \leq r$ , which proves the density of  $D$ .

Since  $D \in M$  is dense and  $G$  is generic we have  $G \cap D \neq \emptyset$ . Let  $\bar{p} \in D \cap G$ . Since  $p_n \in G$  for all  $n < \omega$  it must be  $\bar{p} \leq p_n$  for all  $n < \omega$ . So  $\bar{p}$  is the element we were looking for.

Now consider  $B = \{n : \bar{p} \Vdash \check{n} \in \dot{A}\}$ . Note that  $B \in M$  since one can use the equivalent internal definition of forcing. We claim  $B = A$ .

- $A \subseteq B$ :  $n \in A \Rightarrow \exists r \in G(r \Vdash \check{n} \in \dot{A}) \Rightarrow \bar{p} \Vdash \check{n} \in \dot{A}$  because otherwise  $\bar{p}, r \in G$  would be incompatible.
- $A \supseteq B$ :  $n \in B \Rightarrow \bar{p} \Vdash \check{n} \in \dot{A} \Rightarrow V[G] \models n \in A$ .

This means that  $\mathcal{P}(\omega)^V = \mathcal{P}(\omega)^{V[G]}$ . Now observe that  $|2^\omega|^V \geq |2^\omega|^{V[G]}$  and since  $|2^\omega|^V \in \lambda$  we get  $|2^\omega|^{V[G]} \in \lambda$ . Furthermore from  $V[G] \models \lambda = \omega_2$  we get  $V[G] \models |2^\omega| < \omega_2$ . Since  $V[G] \models ZFC$  and  $|2^\omega| \geq \omega_1$  is a theorem of ZFC, we also get  $V[G] \models |2^\omega| \geq \omega_1$ . So we conclude  $V[G] \models |2^\omega| = \omega_1$ . □

For the next problem, recall that  $Add(\omega_1, 1)$  is the poset to add one Cohen subset of  $\omega_1$ . More precisely,  $Add(\omega_1, 1) = \{p : \omega_1 \rightarrow 2 : |p| \leq \omega\}$ , ordered by reverse inclusion.

**Esercizio 2.** Suppose that  $G$  is  $Add(\omega_1, 1)$ -generic over the ground model  $V$ . Show that  $V[G] \models CH$ .

*Proof.* As in the previous exercise, observe that  $(2^\omega)^V = (2^\omega)^{V[G]}$ ,  $(\omega_1)^V = (\omega_1)^{V[G]}$  because  $P$  is  $\omega_1$ -closed. Now let  $g = \bigcup G \in V[G]$ . Then  $g$  will be the characteristic function of a subset of  $\omega_1$ . Let's call it  $A$ . Now consider  $f : \omega_1 \rightarrow \mathcal{P}(\omega), \alpha \mapsto B_\alpha$ , where  $n \in B_\alpha \iff \alpha + n \in A$ . We claim  $f$  surjective. In fact let  $B \subseteq \omega$ . We want to find  $\alpha < \omega_1$  such that  $B_\alpha = B$ . Equivalently, we want to prove that  $\{\alpha < \omega_1 : \forall n < \omega (n \in B \iff \alpha + n \in A)\} \neq \emptyset$ . Consider

$D_B = \{p \in P : \exists \alpha < \omega_1 \forall n < \omega (\alpha + n \in \text{dom}(p) \wedge (p(\alpha + n) = 1 \iff n \in B))\}$ . Then

$D_B \in M$  and  $D_B$  is dense by regularity of  $\omega_1$ . Let  $r_B \in D_B \cap G$ . Then  $r_B \Vdash \dot{B} \in \text{ran}(\dot{f})$ , which means that  $V[G] \models f$  surjective. Equivalently,  $V[G] \models CH$ . □